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Review of Large Language Models for Patient and Caregiver Support in Cancer Care Delivery

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ABSTRACT

This narrative review examines the current landscape and evidence regarding large language model (LLM) applications designed to support patients with cancer and caregivers. We analyzed peer-reviewed literature, conference proceedings, and implementation studies exploring LLM use in oncology patient support. Applications cluster in four primary domains: education and information delivery, symptom checking and triage, telehealth integration, and clinical trial participation. Studies demonstrate promising accuracy for basic cancer information delivery, although performance varies for complex clinical scenarios. Early research shows preclinical feasibility and acceptability of LLM-enhanced tools for patients, but effectiveness data remain limited. Implementation barriers include scalable monitoring, equitable access, maintaining privacy standards, and validating accuracy across diverse populations. We also examine potential future applications across the cancer care continuum, from prevention through end-of-life care, and propose strategies for development and implementation. Additionally, we present a framework to guide physician-patient discussions regarding LLM use in oncology, addressing privacy concerns, setting appropriate expectations, and ensuring safe integration into care delivery. Future research should use robust evaluation frameworks focused on safety and patient-centered outcomes while carefully considering health equity implications. As these technologies evolve, maintaining focus on evidence-based validation will be crucial for realizing their potential to enhance cancer care delivery, engagement, and patient satisfaction.

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INTRODUCTION

Large language models (LLMs) have emerged as promising tools to address critical gaps in health care.[1,2] These artificial intelligence (AI) systems process and generate human-like text through sophisticated pattern recognition, learning from vast collections of text data to understand context, generate coherent responses, and engage in natural dialogue—capabilities that distinguish them from traditional rule-based systems.[3,4] Recent analyses indicate that over 80% of health care organizations are exploring or implementing LLM solutions, yet their application to direct patient support remains limited.[5] Patients with cancer and their caregivers face particularly complex information needs throughout their prolonged care journeys. Studies show that nearly half of the patients with cancer report unmet informational needs during treatment, and approximately 70% of caregivers indicate they feel inadequately prepared for their care responsibilities.[6-10] These gaps in support between clinical encounters represent a critical opportunity for LLM applications.

Research on LLM applications in oncology has expanded significantly, with both clinicians and patients adopting these systems for health information needs. Online discussion forums reveal growing patient utilization of LLMs for self-directed health care activities, including symptom assessment, self-education, and medical decision-making support.[11-13] However, their potential impact on patient and caregiver experience remains poorly characterized.[14] Current applications focus primarily on clinical documentation, administrative tasks, and clinical decision support, leaving considerable uncertainty about their role in patient-facing contexts.[14-18] Several recent reviews have examined LLMs in oncology from different perspectives, including patient-facing applications (Raynaud et al), technical categorizations with emphasis on safeguards (Zitu et al), head and neck cancer–specific applications (Lacerda de Souza et al), and broad evidence overviews (Iannantuono et al).[14,19-21] Building upon this foundation, this review uniquely focuses on implementation science and care delivery integration for patient and caregiver support. We examine the current landscape of LLM applications designed to support cancer patients and caregivers, highlighting both promises and pitfalls, examining implementation challenges, evidence of effectiveness, and stakeholder engagement strategies. We also identify critical knowledge gaps and propose future research directions, including a practical AI discussion

checklist to guide patient-provider conversations about LLM use in oncology care

METHODS

We conducted a comprehensive literature search using PubMed, Google Scholar, and arXiv databases to identify studies examining the use of LLM, AI, and chatbots in oncology care. The search included articles published through January 2025. Search terms combined technology-focused keywords ("large language models," "artificial intelligence," and "chatbots") with oncology-specific terms ("cancer," "oncology," "patient experience," "caregivers," and "digital health"). This search was supplemented by reviewing conference proceedings and reference lists of relevant publications.

The identified studies represented diverse methodological approaches reflecting the early stage of this field. Study types included early evaluations and preliminary feasibility assessments, preclinical simulations, pilot studies, pragmatic trials, and limited randomized controlled trials (RCTs). Most studies represented early-phase investigations rather than full-scale clinical implementations, with many focused on proof-of-concept demonstrations or small-scale pilot testing. This heterogeneity in study maturity and design reflects the rapidly evolving nature of LLM applications in oncology and the predominance of exploratory rather than confirmatory research in this emerging field.

OVERVIEW OF LLM APPLICATIONS

Despite the exponential growth of AI applications in health care, our review revealed a scarcity of studies exploring LLM applications aimed specifically at patients and caregivers in oncology. Yet, surveys indicate patients are using LLMs for health information more broadly.[22] Patients and caregivers have increasingly shared their experiences with LLMs through social media platforms and digital publishing outlets, describing how they leverage these tools to navigate their healthcare journeys.[23-25] Although these anecdotal experiences do not meet traditional standards of clinical evidence, they highlight a significant disconnect between formal research and the full extent of real-world LLM adoption by those affected by cancer. This decentralized, grassroots use makes it challenging to gauge the complete scope of dissemination and the myriad ways these tools are being integrated into daily lives.

Another major challenge in evaluating LLM applications is the rapid pace of model development. Traditional publication timelines, typically requiring 6-12 months for peer review, cannot keep up with the frequent iterations of major LLM architectures. Consequently, published evaluations often lag behind the current capabilities of these models. This issue is further compounded by the absence of standardized evaluation frameworks for health care applications. Researchers use diverse methods—from varying prompt structures to inconsistent evaluations—which makes cross-study comparisons difficult. Additionally, regular updates to commercial LLMs can significantly alter their behavior, complicating efforts to establish reliable performance benchmarks or reproduce published findings. Despite these challenges, our review strives to provide a comprehensive picture of the current state of LLM applications in patients with cancer and caregiver support.

In the following section, we summarize the current landscape of LLM applications in oncology and organize our findings around four key themes: education and information delivery, symptom checking and triage, telehealth and virtual care, and research and clinical trials.

Education and Information Delivery

LLM-based chatbots show particular promise in providing evidence-based information. Görtz et al[26] examined a chatbot designed for men at risk for prostate cancer, with participants strongly endorsing increased chatbot utilization for information delivery. Similarly, Halawani et al[27] compared AI-generated educational content against standard patient materials for kidney cancer, demonstrating that LLMs could produce accurate and readable information, although noting potential issues with content complexity.

Evidence regarding LLMs' ability to generate reliable patient-facing cancer information remains mixed and highly dependent on study methodology. Early evaluations of Generative Pretrained Transformer (GPT)-3.5 found that although responses to cancer-related queries were largely accurate, they frequently contained interweaved misinformation about treatments for breast, prostate, and lung cancers.[28] For breast cancer screening and prevention, Haver et al[29] found that 88% of ChatGPT's recommendations were deemed appropriate by board-certified breast radiologists. The same group reported comparable accuracy rates for lung cancer prevention guidance.[30] However, when examining cervical cancer screening information, researchers found that although LLMs excelled at basic educational content, they performed less effectively with complex clinical scenarios and nuanced cases.[31] Yeo et al[32] evaluated the use of ChatGPT and found that it could offer practical, multifaceted advice for treatment navigation and medical decision making in patients with hepatocellular carcinoma. For caregivers, Sezgin et al[33] assessed multiple publicly available language models' performance in delivering reliable information regarding pediatric oncology with promising results but with variation across models, although these findings represent early-phase investigations. Overall, the evidence on whether LLMs can reliably generate patient-facing cancer information remains mixed, partially because findings vary with different study designs and are often using differing benchmarks.[34] This variability continues to complicate the translation of preclinical evaluations into clinical practice.

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Beyond traditional educational content and clinical guidance, LLMs could potentially leverage extensive patient-generated experiential data from online forums and social media platforms to benefit both health care providers and patients. This real-world evidence complements standard medical information by capturing authentic patient perspectives and practical insights, as demonstrated by Kouzy et al[35] summarizing the patient's sentiment regarding cervical brachytherapy. For providers, similar analyses of online forums offer windows into patient experiences outside clinical encounters, for example, educating providers on cancer concern discussions and glucagon-like-peptide agonist medication.[36]

These preliminary evaluations suggest that LLM-based educational tools remain in the early stages of development, necessitating consistent and reliable evaluation methods—particularly for more complex cases. Current limitations largely reflect the rapidly evolving nature of LLM technology, with most studies focusing on preclinical question-answer capabilities rather than comprehensive clinical validation in real patient care. Research indicates that future iterations may benefit from enhanced reasoning frameworks[37,38] (ie, structured processes that facilitate multistep logical analysis and improve interpretability) and agentic pipelines[39-41] (ie, orchestrated sequences of specialized LLM agents working collectively on distinct tasks) to optimize both accuracy and personalization of information delivery.[28]

Symptom Checking and Triage

LLM-assisted symptom checkers represent an evolution from traditional digital health tools, leveraging LLMs' dialogue capabilities to conduct clinical interviews and symptom assessments. Although digital symptom checkers have existed since the early 2000s, LLM-enhanced versions offer more sophisticated, conversational interactions that could support patient self-triage and care navigation.[42]

This capability is particularly relevant in oncology, where patients must monitor for critical symptoms that may require urgent medical attention, such as febrile neutropenia or treatment-related complications.[43] Zhu et al[44] evaluated a retrained language model designed to interact with patients with head and neck cancer receiving radiation therapy. Using a small, retrospective data set of patient questions, the model demonstrated high accuracy in providing supportive care recommendations and managing treatment-related symptoms, suggesting potential utility for routine symptom monitoring.

The integration of LLMs with electronic patient-reported outcomes (ePROs) represents another promising application. Liao et al[45] examined LLMs' ability to analyze side-effect data from ePROs in patients with cancer, finding that LLMs could effectively identify trends and generate clinically relevant insights. Wu and Bibault[46] tested two GPT4-driven approaches for capturing patient-reported toxicities during radiotherapy treatment visits preclinically, demonstrating how structured digital collection of symptoms could enhance patient-physician communication while reducing documentation burden.

A key consideration in implementing these tools is ensuring accessibility across diverse patient populations. These efforts are still in their early stages, and more implementation and validation studies are needed. Whether LLM-enhanced symptom checking and triage tools have any measurable effect on patient engagement or outcomes remains to be thoroughly studied, particularly regarding ongoing symptom monitoring and management between clinical visits.

Telehealth and Virtual Care

Telehealth adoption in oncology care accelerated during the COVID-19 pandemic, creating opportunities for LLM integration in virtual care delivery.[47,48]

Recent studies have demonstrated promising LLM applications in virtual care settings. Cho et al[17] evaluated GPT-4 with fusion prompting for oncology telephone consultation analysis, showing effective classification of patient intent for treatment, scheduling, and urgent care routing. The system demonstrated particular utility in streamlining navigation through complex health care systems, with high accuracy in directing patients to appropriate resources.

Recent investigations have explored the integration of LLMs in patient portal communications, with emerging evidence suggesting promising outcomes.[49,50] In the context of digital health communication, LLMs present a novel avenue for enhancing patient-provider interactions. A simulation study examining oncology-related patient inquiries demonstrated that LLM implementation could enhance educational content quality, although it identified potential risks of harmful errors because of inadequate conveyance of urgency and poor generalization to issues common in patients with cancer, emphasizing the necessity of human oversight.[50] Recent work by Hao et al[49] corroborated these findings, showing that although LLM-powered messaging systems could generate personalized patient responses, human clinical supervision is needed. This human-in-the-loop approach proved particularly crucial as health care providers consistently delivered more actionable guidance compared with automated responses in both studies. Although LLMs appear to provide more comprehensive disease education, the optimal synergy between human expertise and artificial intelligence remains to be fully elucidated. Current evidence suggests this technological integration merits further investigation, particularly regarding implementation strategies and clinical validation.[51] Although similar applications have been documented in other medical domains, the extrapolation of these findings to oncology care requires additional research and validation.[52,53]

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Yang et al[54] investigated virtual cancer care communication through provider interviews, identifying key implementation challenges and opportunities. Their study found strong provider and patient interest in LLM assistants for symptom monitoring and communication support, while noting concerns about accuracy and customization. The system showed particular promise in providing regular patient check-ins, conversation transcription, and mental health support. Zeng et al[55] examined how GPT-4 could enhance virtual language rehabilitation for oncology patients with aphasia, demonstrating that AI-augmented communication tools could provide personalized therapy through a combination of eye tracking and context-aware language generation. Preliminary evaluations indicated that these applications were feasible and acceptable to both patients and providers, although implementation challenges regarding provider availability and the sensitive nature of cancer care interactions require further study.

Research and Clinical Trials

Clinical trial accessibility and participation represent another critical domain in cancer care where LLM applications show promise.[56] Trial matching, a complex process that currently is a barrier to efficient accrual, has demonstrated potential for technological enhancement. Jin et al[57] conducted a landmark study demonstrating the efficacy of LLM-based automation in patient-to-trial matching, showcasing the model's capability to effectively retrieve, predict, and establish trial-patient concordance. Building on this foundation, Datta et al[58] developed a comprehensive end-to-end LLM-assisted trial matching system, which achieved high precision metrics across multiple pathologies, including breast cancer. The utility of LLMs extends beyond matching algorithms into the realm of patient education and engagement. Notably, a clinical pilot revealed that GPT-generated trial summaries, after human clinical review, enhanced patients' comprehension of a trial they were consenting to.[59] These findings collectively suggest that LLM integration could address multiple barriers in clinical trial participation.

FUTURE APPLICATIONS AND CONSIDERATIONS

As LLM technology matures, numerous potential applications emerge to address critical needs in cancer care delivery.[51] Although some applications show early promise, most require careful evaluation to ensure they meaningfully improve patient and caregiver experiences and do not introduce harm (Table 1). Educational support and visit summarization represent particularly promising early applications. For example, LLM-based tools could provide personalized summaries of medical appointments in plain language, helping reduce information overload—a common challenge identified by patients and caregivers.[8] These systems show potential for improving understanding and retention of complex treatment plans, although concerns exist about oversimplification of critical details.[50,60,61]

Care coordination and administrative support tools demonstrate another area of significant potential impact. LLM-assisted systems could help navigate insurance claims, coordinate appointments across multiple specialists, and manage medication schedules.[62,63] Early implementations suggest these tools could reduce administrative burden and improve care continuity, although integration with existing health care systems presents significant technical and regulatory challenges.

Emotional support applications, although promising, require careful safety and ethical consideration. LLM-driven platforms could provide 24/7 psychosocial support and stress management tools, particularly valuable in settings with limited counselor availability.[64] However, these applications cannot fully replicate human connection and risk creating dependence on LLMs without appropriate professional oversight.

The successful implementation of these tools will require robust evaluation frameworks focused on patient-centered outcomes. Traditional metrics may not capture the full impact of these interventions, necessitating new approaches to measuring success. As demonstrated during the COVID-19 pandemic, patients often adopt digital health solutions more rapidly than health care systems, highlighting the importance of careful implementation strategies that balance innovation with patient safety and clinical efficacy.[65]

Table 1 summarizes key potential applications, providing examples of implementation approaches and associated challenges that warrant consideration as the field evolves.

MEASURING THE EFFECTIVENESS OF LLM INTERVENTIONS

The evaluation of LLM interventions in oncology requires both scientific rigor and meaningful patient engagement in defining and assessing outcomes. The field currently lacks standardized metrics and methodologies for measuring effectiveness.[66] This gap underscores the importance of incorporating diverse stakeholder perspectives, particularly those of patients, to ensure evaluation methods align with meaningful clinical and personal outcomes.

Study Design Considerations

Although RCTs remain foundational for establishing causality, the rapid evolution of AI technology presents unique challenges for traditional study designs.[67] LLM capabilities may advance significantly during the time required to complete and publish an RCT, potentially limiting the relevance of findings. Some interventions may demonstrate such clear benefits that requiring RCT-level evidence could unnecessarily delay implementation. In these cases, evidence from pilot studies, user feedback, or pragmatic trials could provide sufficient justification for adoption.[68] However, as there remains many unknowns about the actual risks

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TABLE 1. Artificial Intelligence Applications for Patient Engagement and Support Throughout the Cancer Care Continuum

Use Case	Care Continuum Phase(s)	Implementation Example	Potential Advantages	Potential Disadvantages
Information assessment	P, ED, D, T, S, PC, EoL	Automated verification of cancer-related medical information	Reduces misinformation spread; provides evidence validation; enables rapid fact-checking	Introduces algorithmic bias; requires trust building; demands data verification
Clinical documentation support	ED, D, T, S, PC, EoL	Automated generation of plain-language visit summaries	Enhances patient comprehension; personalizes information delivery; creates consistent documentation	Risks oversimplification; may introduce transcription errors; requires physician verification
Language translation services	ED, D, T, S, PC, EoL	Automated translation of clinical communications	Broadens care accessibility; reduces communication barriers; accelerates information delivery	Misses cultural nuances; decreases translation accuracy; requires human verification
Digital symptom management	T, S, PC	Automated assessment and triage of symptoms	Provides continuous monitoring; enables early intervention; standardizes assessment	Generates alert fatigue; requires technical literacy; lacks human context; contributes to digital toxicity
Treatment schedule coordination	T, S	Digital platform for tracking medical events	Facilitates coordination; enhances compliance; reduces scheduling errors	Demands EHR updates; requires system integration; increases technical overhead
Medication compliance system	T, S, PC	Digital platform for medication monitoring	Improves adherence; facilitates toxicity detection; automates tracking	Creates notification fatigue; requires data protection; demands maintenance
Clinical trial identification	P, ED, D, T, PC	Automated screening for trial eligibility	Expands trial access; standardizes screening; accelerates recruitment	Requires data accuracy; needs frequent updates; misses clinical nuances
Insurance process management	ED, D, T, S, PC	Automated claims processing system	Reduces administrative burden; identifies coverage gaps; accelerates reimbursement	Requires policy updates; demands security measures
Financial navigation support	D, T, S, PC	Digital platform for cost management	Provides cost transparency; facilitates assistance; enables planning	Requires data accuracy; demands regular updates
Care network coordination	ED, D, T, S, PC, EoL	Digital system for patient support management	Enhances resource allocation; improves coordination; facilitates tracking	Requires engagement; demands privacy protection
Patient transportation coordination	ED, D, T, S	Automated medical transport system	Reduces delays; optimizes resources; enhances accessibility	Demands integration
Pharmacy management	T, S, PC	Digital medication management system	Optimizes access; identifies alternatives; manages prescriptions and cost	Requires integration; demands updates; needs compliance
Remote care management	T, S, PC	Virtual platform for monitoring patients	Extends specialist reach; reduces travel; enables monitoring	Requires rigorous pilot studies for noninferiority or clinical gaps
Patient support connection	D, T, S, PC	Digital peer-matching platform	Facilitates support; enables connection; enhances experience	Creates verification challenges
Advanced care planning	PC, EoL	Digital platform for care planning	Personalizes planning; enhances documentation; supports sensitivity	Lacks emotional context; requires oversight; risks miscommunication; misses clinical, cultural, and ethical nuances
Home-based hospice support	PC, EoL	Remote monitoring platform	Extends care quality; reduces ER visits; supports decisions	Requires infrastructure; demands monitoring
Disability documentation	T, S	Automated leave paperwork system	Streamlines documentation; reduces burden; ensures compliance	Requires verification
Patient documentation system	T, S, PC	Digital symptom tracking platform	Standardizes recording; enhances communication; facilitates tracking	Introduces privacy issues; requires monitoring of noise:signal ratio
Psycho-social support	ED, D, T, S, PC, EoL	Individualized therapy system	Increases access; enhances coping with stress and change; identify intervention needs	Lacks emotional context; requires oversight; needs clinical validation

NOTE. This table is not meant to be comprehensive of all potential LLM uses, but instead to provide an overview of potential future applications throughout the cancer care continuum. Abbreviations: D, diagnosis; ED, early detection; EHR, electronic health record; EoL, end-of-life care; ER, emergency room; LLM, large language model; P, prevention; PC, palliative care; S, survivorship; T, treatment.

and benefits of LLMs, implementation without robust clinical evidence should be limited to low-risk applications and those where there is an expert-in-the-loop to identify and rectify errors before they reach the patient.

Pragmatic trials offer particular value by evaluating LLMs in routine clinical settings and capturing real-world factors such as workflow integration and diverse patient populations.[69,70] These studies acknowledge the complexity of oncology practice while providing insights into feasibility, scalability, and patient acceptance. Pilot studies serve as essential preliminary steps, allowing researchers to assess usability, understand the impact of human factors, refine outcome measures, and identify potential challenges before launching larger-scale investigations.[71,72]

Defining and Measuring Effectiveness

The definition of effectiveness must extend beyond traditional LLM evaluation metrics such as accuracy and precision.[66] Although these remain important for clinical applications, they cannot capture crucial dimensions such as trust, emotional well-being, or quality of life. Early patient and caregiver engagement in study design as well as implementation scientists can lead to tailoring of effectiveness measures.[73] Among these are patient-reported outcomes using validated instruments to measure anxiety, distress, and satisfaction, as well as clinical endpoints such as treatment adherence and emergency department utilization—both requiring careful study design to establish causality with LLM interventions.[67,68] Finally, engagement metrics, such as usage patterns and retention rates, offer insights into real-world utility and adoption.[71] Low engagement may indicate design flaws, while sustained use suggests meaningful value.

Methodological Considerations

Although expanding effectiveness definitions to include patient perspectives is essential, maintaining scientific rigor remains paramount. Studies should use designs that enable causal inference, ensure reproducibility, and minimize bias when ethically and pragmatically feasible.[69] Novel approaches such as stepped-wedge designs or rolling enrollment can balance methodological robustness with the dynamic nature of LLMs in oncology.[70,74,75]

Patient Partnership in Evaluation

Given the heterogeneity of cancer care experiences, ongoing patient involvement throughout the intervention's lifecycle is crucial.[76,77] Patient participation in advisory boards, study design, and qualitative research can identify unexpected challenges and refine success metrics to better reflect clinical realities.[71,78] This collaborative approach ensures that effectiveness measures evolve alongside LLM capabilities while maintaining relevance across diverse clinical contexts.

CLINICAL INTEGRATION, ETHICS, AND FUTURE DIRECTIONS

The successful implementation of LLM tools in oncology extends beyond technical performance. Although measuring tool effectiveness remains crucial, successful integration requires attention to the broader implementation process, taking into account the context of existing clinical workflows, the engagement and buy-in of key stakeholders, and sensitivity to the underlying ethical principles surrounding its use.[79] When integration efforts do not meet initial expectations, it is important to assess whether the shortfall stems from the intervention's ineffectiveness or from suboptimal implementation. In this context, clearly defining and measuring implementation outcomes throughout the process is necessary for successfully translating LLM tools from development settings into real-world clinical environments.[80] This section outlines potential implementation strategies and outcomes of interest when integrating LLM tools, and presents a practical AI discussion checklist to guide provider-patient conversations about LLM tool usage in cancer care.

Clinical Integration

Embedding LLM outputs into electronic health records or patient portals can streamline clinical workflows and facilitate real-time data sharing among care teams. Preliminary pilot studies suggest LLM-generated summaries of patient encounters and automated symptom management prompts may reduce documentation time.[46] However, these efficiency gains risk might be offset by verification fatigue, where providers must validate every piece of LLM-generated content for accuracy.[81]

Implementation strategies should focus on adaptability while preserving fidelity to mitigate validation burdens.[82] Selective, task-specific AI adoption, such as using LLMs for summarizing patient-generated queries rather than fully automating progress notes or clinical decision support, exemplifies this approach. Clinicians are thus positioned as AI supervisors, leveraging efficiency while maintaining oversight and increasing acceptability.[78] Clear documentation will be an essential part of the process, with disclaimers indicating LLM assistance to delineate accountability.

Successful clinical integration also requires systematic post-deployment monitoring including bias audits, performance tracking, and feedback mechanisms to detect model drift and evolving hallucination patterns.[83] These continuous monitoring frameworks will ensure sustained accuracy and clinical safety as both the technology and medical evidence evolve over time.[84] Future work is needed to develop new, validated methods for overseeing and monitoring generative LLM output for patient-facing clinical contexts.

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Ethical Considerations

The implementation of LLMs in oncology patient care raises several ethical challenges. A primary concern is that these models may encode and amplify historical biases present in training data, potentially leading to discriminatory outputs that disproportionately affect underserved populations.[85,86] LLMs also present unique privacy concerns, as they require access to sensitive patient information to generate contextually appropriate responses, necessitating robust data protection measures. Of particular concern is the growing use of publicly available LLMs (such as ChatGPT) by patients and caregivers seeking medical information—these platforms are not Health Insurance Portability and Accountability Act-compliant, may store and repurpose entered information for model training, and offer no guarantees of data privacy or security. The black-box nature of many LLM systems raises transparency issues, particularly regarding the quality and comprehensibility of their outputs to patients and caregivers. Additionally, LLMs can produce convincing but factually incorrect information (hallucinations), which is particularly dangerous in oncology where misinformation could significantly affect treatment decisions and outcomes. These errors can be particularly difficult to detect because LLM output is often very convincing, sycophantic, and authoritative.[87,88] These transparency limitations raise critical concerns about patient autonomy and informed decision making. Patients may overrely on AI-generated information without understanding how responses were generated or recognizing inherent limitations, potentially compromising their ability to make truly informed health care decisions. Clear guidance on LLM limitations and appropriate safeguards are essential to preserve patient autonomy while enabling beneficial use. These challenges underscore the need for careful oversight, transparency, patient and clinician education, and clear accountability and responsibility frameworks.

Health Equity and Access

Although LLMs offer potential benefits for improving health care engagement in cancer care, significant implementation barriers exist. Patients from low socioeconomic backgrounds often lack reliable access to the necessary technology infrastructure, including consistent internet connectivity and devices.[89] Non-English speakers and those with limited technological proficiency face additional challenges in navigating digital health platforms.[90] These disparities risk creating a two-tiered system where advanced LLM tools are only available in well-resourced environments. To address these challenges, health care organizations must invest in comprehensive infrastructure and training programs that target both patients and health care workers. This dual approach should focus on developing an AI-ready workforce capable of effectively implementing and overseeing these technologies while ensuring equitable patient access. Furthermore, health care systems must establish robust support structures, such as digital navigators or community partnerships, to facilitate technology adoption among vulnerable populations. Regular monitoring of access patterns, user demographics, and outcomes remains essential to identify and address disparities in technology utilization. Success in implementing LLM-based tools in oncology care will ultimately depend on our ability to ensure these advances benefit all patients, regardless of their socioeconomic status or technological proficiency.

Patient Engagement and Education

Many patients already use public LLM platforms (eg, ChatGPT) for medical queries. Surveys indicate that patient use of AI for health information is increasing regardless of formal clinical guidance, creating an urgent need for proactive rather than reactive engagement by health care providers. Rather than discourage this practice, clinicians can support patient autonomy while explaining privacy risks—particularly that personal health information entered into public platforms may be repurposed for model training or stored by third parties. This guidance should extend beyond individual clinical encounters to systematic patient involvement in LLM tool development.

Patient advisory committees play a crucial role in shaping LLM implementation. These committees can help identify potential barriers to adoption, suggest refinements to user interfaces, and ensure AI tools address real patient needs rather than perceived ones. This engagement underscores the importance of identifying key stakeholders early in the implementation process. By incorporating the perspectives of patients and caregivers, particularly through collaborative models, health care systems can better align the development of LLM tools with the priorities that matter most to users.[91] Early experiences suggest that tools codesigned with patients achieve higher engagement and satisfaction rates than those developed solely by technical teams.[92]

The AI discussion checklist, developed with input from a patient and an implementation researcher, serves as an initial framework to facilitate health care provider-patient communication regarding LLM implementation in clinical care. Although this review specifically addresses LLM technologies, the checklist uses the broader term "AI" to align with general patient understanding and recognition. The framework (Fig 1) introduces key discussion domains: AI disclosure protocols, information credibility assessment, data privacy considerations, and the integration of AI-derived insights into clinical team communications.

This checklist (Fig 2) functions primarily as a conversation catalyst rather than a comprehensive clinical tool. It establishes foundational communication elements that preserve patient autonomy while promoting appropriate AI tool

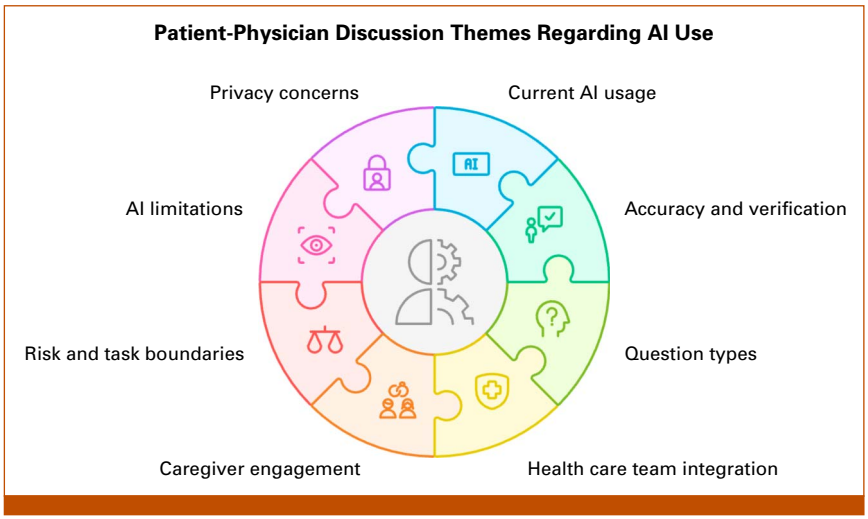

FIG 1. Infographic on themes to guide patient-physician communication regarding AI use in oncology. For a checklist to guide AI-related discussions with patients in the clinic, refer to Figure 2. AI, artificial intelligence.

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utilization within clinical workflows. Although these questions are general in nature, they can be adapted to specific cancer types and treatment contexts as needed. Health care organizations should view this framework as a starting point for developing institution-specific protocols, recognizing that individual clinical encounters may necessitate additional discussion points on the basis of specific contexts and patient needs.

The checklist serves as both a preliminary educational resource and operational framework for health care teams implementing AI in clinical practice. Although it enables informed discussions about AI technologies, the rapidly evolving nature of these applications necessitates parallel evolution in workforce training and stakeholder engagement.[93] Educational programs must equip providers to evaluate AI outputs and integrate patient-facing solutions responsibly, while fostering a collegial, power-sharing model that positions clinicians as central stakeholders in the implementation process.[94] This collaborative approach increases the likelihood that providers will view AI-initiated conversations as opportunities for patient empowerment rather than challenges to their expertise, ultimately creating an environment where AI integration becomes both clinically relevant and sustainable over time.

Sustainability and Future Directions

The dynamic nature of oncology—with frequent guideline updates and emerging treatments—necessitates continuous LLM refinement. Models accurate today may become outdated without consistent revalidation. Cross-institutional collaborations can provide diverse data sets for model updating while ensuring cultural adaptability. Integration with wearable devices and telehealth platforms may enable more proactive patient monitoring and early intervention.

From a processes standpoint, it is critical to intentionally incorporate sustainment strategies at the start of the tool's implementation, to ensure long-term success. These strategies may include an emphasis on leadership support, systematic adaptation of the tool to maintain compatibility with the given organization, and ongoing maintenance of workforce skills through training, supervision, and feedback.[95] Furthermore, purposely re-examining the implementation process through continuous evaluation and feedback loops is essential to allow for iterative improvements to both the technology and its integration.[82] These approaches ensure that LLM-based tools will remain effective, relevant, and seamlessly integrated into clinical practices over time.

In conclusion, LLMs offer compelling opportunities to enhance patient and caregiver engagement in oncology through streamlined information delivery and administrative efficiencies, yet their implementation demands careful consideration of potential risks including accuracy concerns, data biases, and privacy challenges. As innovation accelerates, successful integration will require rigorous validation studies, collaborative design frameworks, and proactive policy measures to ensure equitable access across diverse populations. By combining systematic evaluation with participatory development approaches, the oncology community can harness LLMs' transformative potential while safeguarding patient-centered care and addressing existing health care disparities through inclusive training data, multilingual support, and accessibility in resource-limited settings.

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Framework for Patient-Provider Discussions About AI in Clinic

Current AI Usage and Purpose
- Where do you currently look for information regarding your treatment? (eg, Google, social media, support groups)
- Have you tried using any AI (eg, ChatGPT) to ask questions related to your condition, treatment, or test results?
- Which types of questions or tasks did you find most helpful or least helpful (eg, researching side effects, scheduling, symptom management)?

Comfort With Accuracy and Verification
- Do you have any knowledge regarding the reliability of AI tools?
- Are you aware that AI tools might not always provide 100% accurate information and sometimes make things up (so-called hallucinations)?
- How comfortable are you using AI knowing its recommendations may require verification by your care team?

Nature of Questions Asked
- Were you seeking guidance on your treatment plan, emotional support, or broader cancer-related topics (eg, exercise, nutrition)?
- Did the AI address your queries adequately, or did you feel it missed important details?

Collaboration With the Health Care Team
- Have you shared any AI-generated insights or printouts with your oncologist, nurse, or other members of your care team?
- Do you need any help with interpreting or confirming the AI's answers so you can feel more confident about them?
- Would you like to develop a shared framework with your health care team about which types of questions are most appropriate to explore with AI tools versus direct clinical consultation?

Caregiver Involvement
- Is your caregiver also using AI tools on your behalf? If so, how are you coordinating what is asked and which advice you trust?
- Do you or your caregiver have concerns about data privacy, especially if you input personal information into AI platforms?

Task Scope and Risk Tolerance
- What tasks do you feel comfortable letting AI handle—such as listing upcoming appointments or drafting questions for your next clinic visit—versus tasks you prefer a professional to oversee?
- Are there AI-driven suggestions or advice that you found questionable, and how did you resolve those concerns?

Awareness of AI's Limitations
- Have you noticed the AI providing inconsistent or unclear information? Did it ever contradict what was previously discussed with your care team?
- What steps will you take if the AI's information does not align with clinical recommendations (eg, asking the care team for clarification)?

Privacy and Data Handling
- Are you comfortable with the possibility that your questions and health details may be stored or analyzed by third parties if you use public AI services?
- Would you feel more secure if we recommended an AI tool specifically developed or vetted by our institution?

Preferred Level of Evidence
- What information would you need to know about the AI system to feel comfortable using it?

Satisfaction and Engagement
- Has using AI made you feel more in control, more knowledgeable, less anxious, or better prepared for appointments?
- What would help you feel even more at ease—more guidance on using AI, closer collaboration with your care team, or reassurance about data security?
- Do you find the AI interface easy to use, or do you need assistance navigating it?
- Are AI tools available in your preferred language or format?

Future Improvements and Feedback Loop
- How can we best collect your feedback on whether AI tools are addressing your needs?
- Would you be open to participating in surveys or pilot programs to refine and improve AI-based resources for patients in similar situations?

FIG 2. Suggested discussion framework for patient-provider conversations on AI. This nonexhaustive question set offers conversation prompts to explore patient engagement with AI tools. Questions should be tailored to individual patient needs and circumstances. AI, artificial intelligence.

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